R-UNIVERSALITY OF THE SPECIAL UNITARY LIE ALGEBRA OF A HERMITIAN FORM OVER A QUATERNION DIVISION ALGEBRA WITH THE REVERSION INVOLUTION

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Allowed standard prerequisites (editorial).

- The Tits algebra of the natural representation is the quaternion division algebra itself. Used inside the counted proof at source line 663, attributed by the author to page 378 of the cited book KnusEtAl-BookOfInvols. Identifies the single Brauer class whose behaviour over R decides universality.
- The remark that the centre of the symplectic group has order two, so only one Tits algebra occurs. The author's remark labelled OnlyZStar, invoked inside the counted proof at source line 662. Reduces the criterion to that single class.

Source reference. Original remark 2.4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 462–462. Navigation pointer only: the counted proof uses this remark to reduce to a single Tits algebra; its content is declared as a named standard prerequisite.

Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer only: the counted proof ends by invoking the author's equivalence.

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Source reference. Original definition 3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 428–428. Navigation pointer only: the labelled item of that equivalence used at source line 664.

Proposition 0.1. *Let* $\mathfrak{g} = \mathfrak{su}_n(B; D, \tau)$*, where*

- D *is a quaternion division algebra over* $\mathbb{Q}$*,*
- τ *is the reversion anti-involution of* D*, and*
- B *is an invertible* τ*-Hermitian matrix in* $\mathrm{Mat}_{n\times n}(D)$*.*

Then $\mathfrak{g}$ *is* $\mathbb{R}$*-universal if and only if* D *does not split over* $\mathbb{R}$*.*

Proof. Since the center of $\mathbf{Sp}_n$ has order 2, there is only one Tits algebra to consider (see Remark 2.4). This Tits algebra is D (over $\mathbb{Q}$) [4, p. 378], so the desired conclusion is immediate from Lemma 2.1(3). □

References

[1] Demazure M., Grothendieck A., Schémas en groups. III. Structure des schémas en groupes reductifs, Séminaire de géométrie algébrique du Bois Marie, 1962/1964, available at `http://library.msri.org/books/sga/sga/pdf/sga3-3.pdf`.

[2] Harder G., Bericht über neuere Resultate der Galoiskohomologie halbeinfacher Gruppen, *Jber. Deutsch. Math.-Verein.* **70** (1968), 182–216, available at `http://www.digizeitschriften.de/en/dms/img/?PPN=PPN37721857X_0070&DMDID=dmdlog17`.

[3] Kneser M., Lectures on Galois cohomology of classical groups, *Tata Institute of Fundamental Research Lectures on Mathematics*, Vol. 47, Tata Institute of Fundamental Research, Bombay, 1969, available at `http://www.math.tifr.res.in/~publ/ln/tifr47.pdf`.

[4] Knus M.A., Merkurjev A., Rost M., Tignol J.P., The book of involutions, *American Mathematical Society Colloquium Publications*, Vol. 44, Amer. Math. Soc., Providence, RI, 1998.

[5] Lam T.Y., Introduction to quadratic forms over fields, *Graduate Studies in Mathematics*, Vol. 67, Amer. Math. Soc., Providence, RI, 2005.

[6] Morris D., Real representations of semisimple Lie algebras have $\mathbb{Q}$-forms, in Algebraic Groups and Arithmetic, Tata Institute of Fundamental Research, Mumbai, 2004, 469–490.

[7] Platonov V., Rapinchuk A., Algebraic groups and number theory, *Pure and Applied Mathematics*, Vol. 139, Academic Press, Inc., Boston, MA, 1994.

[8] Prasad G., Rapinchuk A., On the existence of isotropic forms of semi-simple algebraic groups over number fields with prescribed local behavior, *Adv. Math.* **207** (2006), 646–660.

[9] Raghunathan M.S., Arithmetic lattices in semisimple groups, *Proc. Indian Acad. Sci. Math. Sci.* **91** (1982), 133–138.

[10] Tits J., Classification of algebraic semisimple groups, in Algebraic Groups and Discontinuous Subgroups (Proc. Sympos. Pure Math., Boulder, Colo., 1965), Amer. Math. Soc., Providence, R.I., 1966, 33–62.

[11] Tits J., Représentations linéaires irréductibles d'un groupe réductif sur un corps quelconque, *J. Reine Angew. Math.* **247** (1971), 196–220.

[12] Tits J., Strongly inner anisotropic forms of simple algebraic groups, *J. Algebra* **131** (1990), 648–677.